

End-to-End Visual Language Navigation with Limited Sensing: A Survey

WEIYE ZHU*, Southern University of Science and Technology, China

MINGMING YU*, Beihang University, China

MIN LIN*, Shenzhen Campus of Sun Yat-sen University, China

HEWEI PAN*, Southern University of Science and Technology, China

TIANTIAN GENG and XIANGCHEN WANG, Southern University of Science and Technology, China

JIAZHAO ZHANG, Peking University, China

BINGQIAN LIN, Shanghai Jiao Tong University, China

MINGFEI HAN and KUNYU WANG, University of Science and Technology of China, China

RONGTAO XU[†], SpatialTemporal AI, China

FENG ZHENG[†], SpatialTemporal AI, China and Southern University of Science and Technology, China

Visual Language Navigation (VLN) aims to enable embodied agents to follow natural language instructions in complex environments. While most existing approaches rely on rich sensor suites, such as panoramic cameras, RGB-D inputs, or LiDAR, and adopt modular pipelines that decouple perception, reasoning, and planning, these assumptions introduce significant challenges for scalability and sim-to-real transfer. This survey revisits VLN under limited sensing conditions, with a particular focus on monocular RGB inputs and continuous spaces that more closely resemble real-world and human-like perception. We emphasize recent advances in large language models (LLMs) and vision-language models (VLMs), which enable end-to-end navigation by directly grounding visual observations in language-conditioned reasoning and action generation, without relying on explicit intermediate representations. We begin by systematically reviewing simulators, datasets, and benchmarks that support standardized research. We then organize existing approaches within a unified, structured framework into four complementary dimensions: *Context Modeling*, *Imaginative Prediction*, *Decision Making*, and *Reflection From Error*. For each dimension, we analyze representative methods, highlighting their design principles and roles in end-to-end navigation under constrained sensing conditions. Finally, we identify open challenges and outline future research directions toward robust, sensor-efficient, and real-world deployable embodied navigation systems. Github repository of this survey: https://github.com/waynechu1021/Awesome_Visual_Language_Navigation

CCS Concepts: • **Computing methodologies** → **Mobile agents**.

Additional Key Words and Phrases: Embodied Intelligence, Vision Language Navigation, Large Language Models, Limited Sensing

*Both authors contributed equally to this research.

[†]Corresponding author.

Authors' Contact Information: Weiye Zhu, Southern University of Science and Technology, Shenzhen, Guangdong, China; Mingming Yu, Beihang University, Beijing, China; Min Lin, Shenzhen Campus of Sun Yat-sen University, Shenzhen, Guangdong, China; Hewei Pan, Southern University of Science and Technology, Shenzhen, Guangdong, China; Tiantian Geng; Xiangchen Wang, Southern University of Science and Technology, Shenzhen, Guangdong, China; Jiazhao Zhang, Peking University, Beijing, China; Bingqian Lin, Shanghai Jiao Tong University, Shanghai, China; Mingfei Han; Kunyu Wang, University of Science and Technology of China, Hefei, Anhui, China; Rongtao Xu, SpatialTemporal AI, Shenzhen, Guangdong, China; Feng Zheng, SpatialTemporal AI, Shenzhen, Guangdong, China and Southern University of Science and Technology, Shenzhen, Guangdong, China.

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Manuscript submitted to ACM

ACM Reference Format:

Weiye Zhu, Mingming Yu, Min Lin, Hwei Pan, Tiantian Geng, Xiangchen Wang, Jiazhaio Zhang, Bingqian Lin, Mingfei Han, Kunyu Wang, Rongtao Xu, and Feng Zheng. 2026. End-to-End Visual Language Navigation with Limited Sensing: A Survey. 1, 1 (May 2026), 37 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Visual Language Navigation (VLN) [3, 79, 80, 99] aims to enable embodied agents to navigate real-world environments by following natural language instructions. As a foundational problem in embodied artificial intelligence, VLN inherently couples perception, mapping, reasoning, and planning. Over the past decade, substantial progress has been achieved through advances in deep visual representations, reinforcement learning, and, more recently, large language models (LLMs) [215, 216, 224, 244]. However, much of this progress has been achieved under strong assumptions about sensor richness and system design. Many existing approaches rely on rich sensor suites, such as panoramic cameras [57, 58, 199], RGB-D sensors [2, 198, 199], odometry [57, 58, 79], LiDAR [31, 63], or explicit metric maps, and organize navigation into modular pipelines that decouple perception, mapping, reasoning, and planning [113, 232]. While effective in benchmarks, these assumptions introduce significant barriers to scalability, real-world deployment, and sim-to-real transfer.

These limitations motivate a renewed focus on navigating with less, particularly under monocular RGB sensing [243]. Monocular cameras are inexpensive, widely available, and more closely aligned with human perception, making them an appealing sensing modality for embodied agents. At the same time, restricting navigation to monocular RGB observations removes access to many explicit intermediate representations commonly used in modular pipelines, such as depth maps [2, 198, 199], panoramic views [58, 79, 199], or explicit metric maps [94, 113, 232]. As a result, maintaining rigid separations between these modules becomes increasingly brittle, and navigation naturally calls for more unified, end-to-end decision-making frameworks.

Recent advances in large language models and vision language models (LLMs/VLMs) [7, 50, 112, 217, 245, 252, 265] provide a promising foundation for this shift. By directly grounding visual observations in language conditioned reasoning and action generation while typically not relying on explicit mapping, these models [204, 242, 243] enable navigation systems in which perception, reasoning, and planning are tightly integrated. Under minimal sensing constraints, such end-to-end formulations offer a principled alternative to traditional hierarchical pipelines. We compared these two different paradigms in Figure 2.

Despite growing interest in VLN, existing surveys [49, 72, 130, 206, 251] primarily focus on general task formulations that combine rich multi-sensor settings with modular design paradigms. Although some recent surveys [99, 129, 251] examine the integration of LLMs/VLMs in VLN, they mainly catalog the diverse roles these models play within existing pipelines, rather than treating them as the core mechanism for end-to-end perception-to-action generation. Consequently, a systematic understanding of VLN approaches that leverage LLMs/VLMs as the central decision-making core remains largely missing, especially under constrained sensing settings.

This survey addresses this gap by systematically reviewing and categorizing recent VLN literature, with a primary focus on end-to-end frameworks that leverage LLMs/VLMs under limited sensing conditions. While we inevitably include methods from broader settings to provide context and completeness, our emphasis remains on this emerging end-to-end paradigm, aiming to present a coherent and focused perspective on its design principles, challenges, and opportunities.

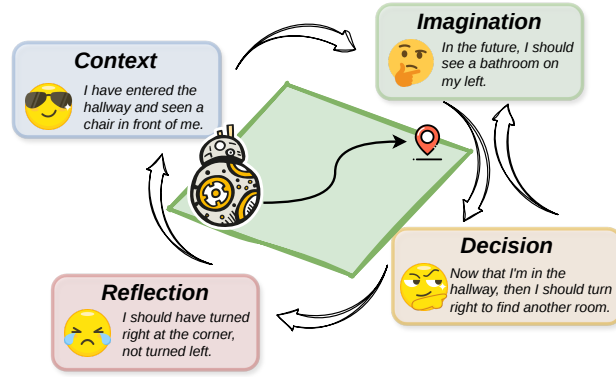


Fig. 1. A unified and structured perspective for visual language navigation. we organize VLN as an end-to-end process within LLM/VLM-based embodied agents. This perspective is structured around four key capabilities: *Context Modeling*, *Imaginative Prediction*, *Decision Making*, and *Reflection From Error*, which jointly capture the information flow of perception, prediction, decision, and adaptation in modern VLN systems.

To structurally organize existing approaches, we identify four key pillars: *Context Modeling*, *Imaginative Prediction*, *Decision Making*, and *Reflection From Error*, as illustrated in Figure 1. Together, these dimensions characterize robust navigation systems, particularly under constrained sensing conditions. Rather than adopting the traditional modular decomposition that separates VLN into loosely coupled components of perception, mapping, reasoning, and planning, we propose a unified and structured perspective that captures the end-to-end information flow from perception to action within LLMs/VLMs-based embodied agent frameworks. Under this perspective, VLN is formulated as a coherent process: the agent maintains and updates visual-temporal context, anticipates and predicts future states, reasons over language instructions, and adapts through uncertainty-aware feedback and error correction. This formulation highlights how modern VLN systems can be systematically understood through these four core capabilities, which jointly integrate perception, foresight, decision, and adaptation.

Overall, this survey aims to provide a unified and structured perspective on instruction-driven visual navigation with limited sensory inputs. Our goal is to clarify the design space for achieving navigation with fewer resources, highlight current limitations and open challenges, and offer guidance for building robust, scalable, and deployable embodied navigation systems. By dissecting representative methods along the proposed dimensions, we distill the design philosophies underlying end-to-end LLM-based VLN approaches and outline promising directions for future research.

The remainder of this survey is organized as follows. Section 2 reviews the VLN task and essential background. Section 3 presents *context modeling*. Section 4 introduces *imaginative prediction*. Section 5 discusses *decision planning*. Section 6 examines *reflection learning from error*. Finally, Section 7 concludes with open challenges and future directions.

2 Preliminaries

In this section, we review the essential background of Visual Language Navigation research and summarize the foundational concepts underlying large language and vision language models (LLMs/VLMs), as well as commonly used simulators, datasets and benchmarks. Our goal is to provide both the necessary technical background of the core components, problem settings, and experimental protocols that underpin VLN research. By consolidating these

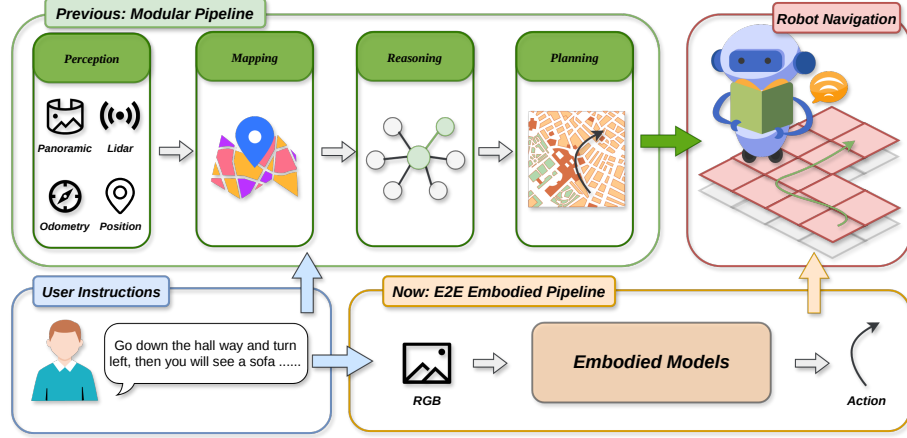


Fig. 2. Comparison between modular and end-to-end paradigms in VLN. Previous approaches decompose navigation into sequential and loosely coupled modules, including perception, mapping, reasoning, and planning, often relying on intermediate representations and multi-sensor inputs. In contrast, modern LLM/VLM-based approaches formulate VLN as an end-to-end process typically without explicit mapping, where perception, reasoning, and planning are jointly realized within a unified framework, enabling tightly integrated information flow.

elements, this section establishes a common conceptual and technical foundation that supports the unified framework adopted throughout this survey.

2.1 Visual Language Navigation (VLN)

A typical VLN task requires an agent to receive a natural language instruction describing the navigation task at the start of navigation. The agent then navigates the environment using egocentric observations. By following the language instruction, the agent outputs a series of control commands to approach the target. Unlike static vision-language tasks, VLN requires agents to reason over sequential observations and actions, integrating visual evidence over time to maintain a coherent internal state. This involves tracking spatial context, handling partial observability, and resolving instruction ambiguities across long-horizon trajectories [69, 91, 98, 175].

Formally, the VLN task can be described as a Partially Observable Markov Decision Process (POMDP) [46]. The agent represents a policy model π^* whose goal is to maximize environmental rewards R by predicting the next action a_t given conditions $C_t = \{o_{1:t}, a_{1:t-1}, \mathcal{I}\}$:

$$\pi^* = \arg \max_{\pi} \sum_{t=1}^T R(\pi(a_t | C_t)), \quad (1)$$

where o_t means the visual observation accessible to the agent at timestep t including panoramic or monocular RGB and depth information, and $\mathcal{I}$ means the task-related natural language instruction from human.

Earlier VLN methods [3, 22, 97, 260, 261] typically represent the environment as a topological graph composed of discrete nodes, where navigation action a_t is formulated as selecting among predefined viewpoints. While such formulations simplify planning and evaluation, they constrain the agent’s action space to discrete, graph-based decisions. In contrast, recent work [23, 204, 242, 243, 254] has increasingly shifted toward navigation in continuous environments, extending the action space to continuous control that more closely reflects real-world embodied navigation. Accordingly, this survey primarily focuses on VLN settings in continuous space.

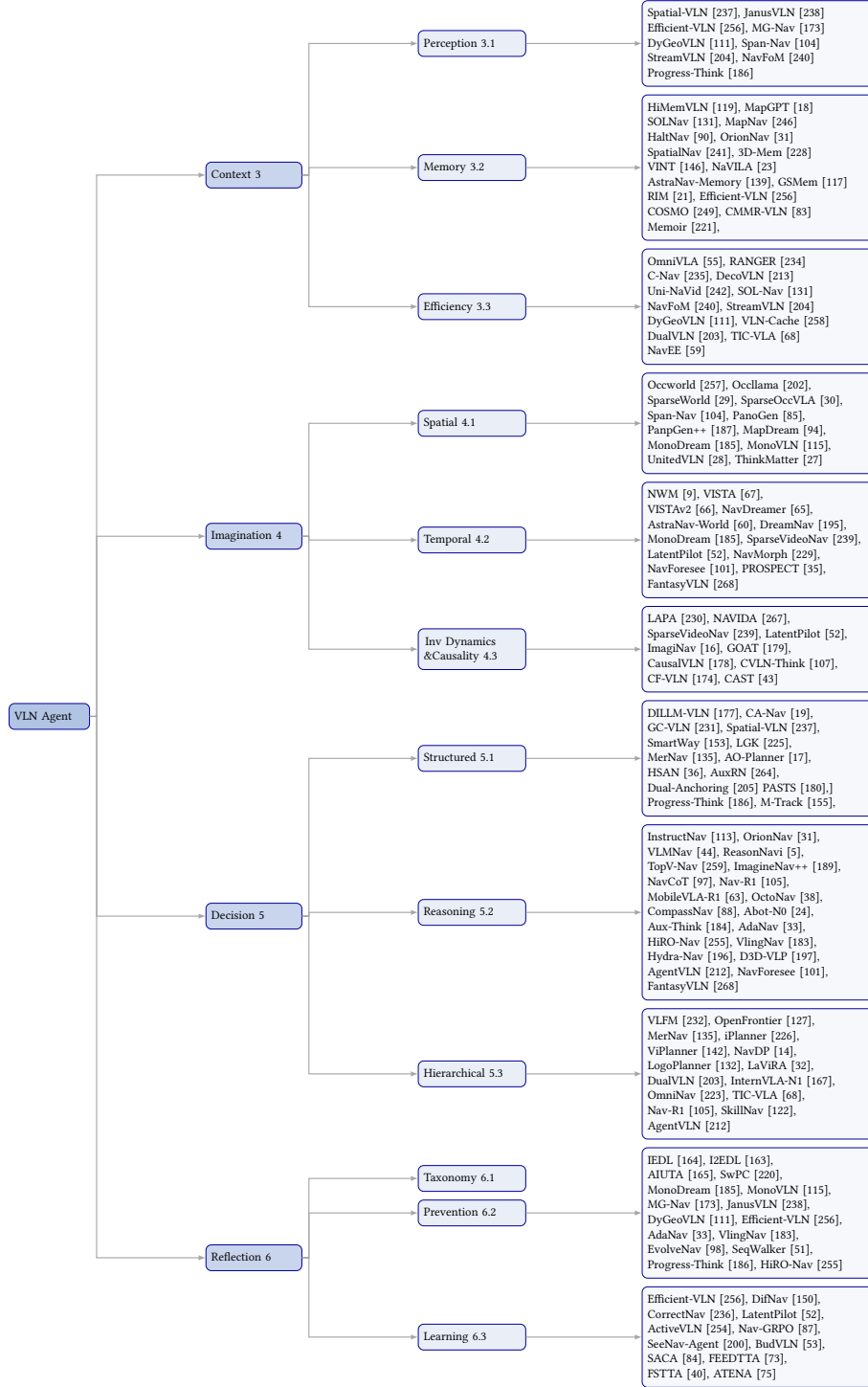


Fig. 3. Taxonomy of this survey and corresponding representative papers.

2.2 Large Language Models (LLMs) and Vision Language Models (VLMs)

Large Language Models (LLMs) [1, 11, 137] arise from breakthroughs in natural language processing, where Transformer-based architectures [172] fundamentally reshaped sequence modeling by enabling strong contextual understanding and compositional reasoning. Vision Language Models (VLMs) [82, 86, 102] extend this paradigm by aligning visual representations with linguistic semantics, allowing models to jointly process perception and language. While such multimodal grounding has been extensively studied in static vision language tasks [4, 160], its extension to embodied settings, where language must be grounded in temporally evolving visual observations and physical actions, remains far from fully understood and continues to pose open challenges.

Early VLN approaches used LLMs and VLMs as high-level planners or decision cores operating over structured inputs [97, 260, 261], whereas recent works [204, 242, 243] have begun to move beyond such modular pipelines toward end-to-end formulations, where language and visual models directly map multimodal observations and instructions to navigation actions. This shift reflects a growing emphasis on unified perception–reasoning–planning frameworks, particularly under constrained sensing conditions such as monocular RGB input, and marks a key transition in how LLMs and VLMs are leveraged within VLN systems.

2.3 Simulation, Data and Control

The effectiveness of sim-to-real policies in VLN largely depends on the level of physical realism, where environmental simulation, data generation, and control modeling jointly determine how navigation policies perceive, learn, and act.

2.3.1 Toward More Realistic Simulation. A simulation architecture suitable for modern VLN must push fidelity along two key dimensions: photorealistic rendering, which reduces the perceptual gap, and physically accurate dynamics, which minimizes discrepancies in motion and interaction. Early indoor VLN research [3, 58] primarily relied on the Matterport3D [15] simulator, which reconstructs textured 3D indoor environments from real-world scans and renders panoramic observations from discrete viewpoints. Concurrently, platforms such as AI2-THOR [77] and Gibson [207] laid important groundwork for improving visual fidelity. Building on these efforts, the Habitat [144] series became the dominant platform for VLN-CE [79, 204, 243, 254, 267], offering highly optimized rendering pipelines and integrating physics engines like Bullet [26] to support rigid-body interactions. By Habitat 3.0 [134], the simulator not only incorporated large-scale real-world datasets such as HM3D [138] and Matterport3D [15], but also introduced realistic sensor noise models, enhancing the realism and unpredictability of embodied perception and control. More recent datasets and platforms, such as SAGE-3D [125], NavGSim [103], Habitat-GS [208] and Isaac Sim, further improve visual realism through techniques like 3D Gaussian splatting [71], narrowing the gap between simulated and real-world visual observations.

To enable full sim-to-real transfer, simulators must also model embodiment and dynamics. Platforms such as VLNVerse [100] and VLN-PE [181] explicitly support parametric robot kinematics, continuous control, collision dynamics, and rich physical interactions, and provide a unified pipeline for data collection, training, and evaluation across diverse robot morphologies, including wheeled, quadrupedal, and humanoid agents.

2.3.2 Towards More Diverse Datasets. Building upon these simulation platforms, various VLN datasets and benchmarks have been proposed to address diverse application scenarios and navigation challenges. As summarized in Table 1, we categorize existing VLN benchmarks along several key dimensions, including application scenarios (indoor or outdoor), underlying simulation platforms (e.g., AI2-THOR, Habitat, Unreal Engine, Isaac Sim), instruction properties (single- or

Table 1. Overview of Vision-and-Language Navigation Datasets and Benchmarks.

Name	Domain	Environment	Inst. Turns	Inst. Type	Inst. Anno	Traj. Anno	Agent Type	Action Space
VLN-CE [79]	Indoors	Habitat	Single	Fine-grained	Human	Planner	Wheeled	Discrete
Robo-VLN [69]	Indoors	Habitat	Single	Fine-grained	Human	Planner	Wheeled	Continuous
ALFRED [154]	Indoors	AI2-THOR	Single	Fine-grained	Human	Planner	Wheeled	Discrete
TEACH [128]	Indoors	AI2-THOR	Multi	Freeform Interactive	Human	Human	Wheeled	Discrete
DialFRED [41]	Indoors	AI2-THOR	Multi	Restricted Interactive	Human/Rule	Planner	Wheeled	Discrete
R2RIE-CE [164]	Indoors	Habitat	Single	Fine-grained	Human	Planner	Wheeled	Discrete
IR2R-CE [78]	Indoors	Habitat	Multi	Fine-grained	Human	Planner	Wheeled	Discrete
HAZARD [262]	Indoors	ThreeDWorld	-	-	-	-	Wheeled	Discrete
HA-VLN 2.0 [34]	Indoors	Habitat	Single	Fine-grained	Human/LLM	Planner	Wheeled	Discrete
SH-IR2R-CE [51]	Indoors	Habitat	Single	Fine-grained	LLM	Planner	Wheeled	Discrete
LH-VLN [157]	Indoors	Habitat	Single	Fine-grained	LLM/Rule	Planner	Wheeled	Discrete
OctoNav [38]	Indoors	Habitat	Single	Fine-grained	Human/LLM	Planner	Wheeled	Discrete
Nav-AdaCoT-2.9M [183]	Indoors	Habitat	Single	Fine-grained	LLM	Planner	Wheeled	Continuous
VLN-PE [181]	Indoors	Isaac Sim	Single	Fine-grained	Human/LLM	Planner	Wheeled/Legged	Hybrid
InternData-N1 [25]	Indoors	Habitat/Isaac Sim	Single	Fine-grained	LLM	Planner	Wheeled/Legged	Hybrid
VLNVerse [100]	Indoors	Isaac Sim	Hybrid	Hybrid	Human/LLM	Planner	Wheeled/Legged	Hybrid
NavSpace [227]	Indoors	Habitat	Single	Fine-grained	Human/LLM	Human	Wheeled	Discrete
VL-LN [64]	Indoors	Habitat	Multi	Freeform Interactive	LLM/Rule	Planner	Wheeled	Discrete
CoIN-Bench [165]	Indoors	Habitat	Multi	Freeform Interactive	-	-	Wheeled	Discrete
IndoorUAV [110]	Indoors	Habitat	Single	Fine-grained	LLM	Human	Aerial	Discrete
UrbanNav [124]	Outdoors	Youtube	Single	Fine-grained	Human/LLM	VO	Legged	Continuous
Talk2Nav [171]	Outdoors	Google Street View	Single	Fine-grained	Human	Planner	-	Discrete
LSCD [159]	Outdoors	CARLA	Single	Fine-grained	Human	Planner	Vehicle	Discrete
CDNLI [140]	Outdoors	CARLA	Multi	Fine-grained	Human/Rule	Planner	Vehicle	Discrete
SDN [123]	Outdoors	CARLA	Multi	Freeform Interactive	Human	Human	Vehicle	Hybrid
Waymo-E2E Driving [162]	Outdoors	Real World	Single	Coarse-grained	-	-	Vehicle	Continuous
Sekai [93]	Outdoors	Real World/UE	Single	Fine-grained	LLM	Visual SLAM	Legged/Aerial	Continuous
EmbodiedCity [39]	Outdoors	UE/AirSim	Single	Fine-grained	Human	Human	Vehicle/Aerial	Discrete
AerialVLN [108]	Outdoors	UE4/AirSim	Single	Fine-grained	Human	Human	Aerial	Discrete
CityNav [81]	Outdoors	SensatUrban	Single	Fine-grained	Human	Human	Aerial	Discrete
OpenUAV [191]	Outdoors	UE4/AirSim	Single	Fine-grained	Human/LLM	Human	Aerial	Continuous
OpenFly [42]	Outdoors	UE/AirSim/GTA V/GE	Single	Fine-grained	Human/LLM	Planner	Aerial	Discrete
UAV-ON [210]	Outdoors	UE/AirSim	Single	Coarse-grained	Human/LLM	Planner	Aerial	Discrete
UAV-Flow [190]	Outdoors	Real World/UE	Single	Fine-grained	Human/LLM	Human	Aerial	Continuous
AirNav [13]	Outdoors	SensatUrban/CityRefer	Single	Fine-grained	Human/LLM	Planner	Aerial	Discrete

multi-turn, fine-grained, restricted, or free-form), annotation sources for both instructions (human or LLM-generated) and trajectories (human demonstrations or planner-generated), supported embodied agent types (e.g., wheeled, legged, vehicular, or aerial), and action space design (discrete or continuous, excluding graph-based formulations that deviate from real-world settings).

2.3.3 Towards More Continuous Control. A key challenge in VLN lies in translating high-level navigation decisions into executable robot actions. Even with accurate perception and environment modeling, mismatches between action representations and real-world control can lead to significant performance degradation. Built upon Matterport3D simulator, early VLN approaches [3, 22, 58, 80, 97] formulate navigation over discrete topological graphs, where agents move between predefined nodes connected by edges. At each step, the agent selects a neighboring viewpoint and transitions to the corresponding node. While this formulation simplifies the problem, it relies heavily on static panoramic observations and lacks the continuity required for physically grounded navigation. With the introduction of simulators that incorporate continuous physical feedback [134, 144], VLN has gradually shifted toward more realistic environments [79, 204, 242, 243]. In these settings, navigation is typically modeled using discrete atomic actions (e.g., move forward 50 cm, turn left 15°, turn right 15°, and stop), which partially approximate robot motion and enable limited transfer to real-world systems. However, such abstractions remain overly coarse, ignoring the need for fine-grained control and leading to error accumulation, trajectory instability, and poor adaptability in complex scenarios. To overcome the limitations of coarse discrete action abstractions, advances in modern simulators [100, 181] have enabled a shift from discrete action prediction to continuous and executable motion representations [60, 203, 240], such as

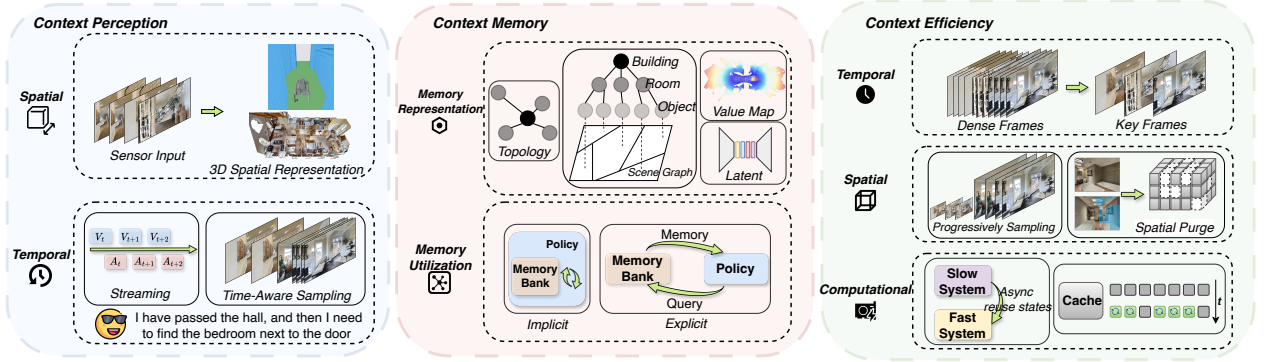


Fig. 4. Categorization of context modeling approaches in VLN. Methods are organized along three key dimensions: *Context Perception*, *Context Memory*, and *Context Efficiency*. Each category summarizes representative approaches that address different aspects of contextual understanding under partial observability, including extracting spatial-temporal and semantic information, maintaining and retrieving historical context for long-horizon reasoning, and improving the efficiency of context representation and reuse.

directly generating dense trajectories or velocity control signals. This transition leads to smoother and more physically consistent behaviors, bringing VLN policies closer to real-world navigation dynamics.

Beyond improving action representations, recent work has begun to optimize the interaction between planning and control. By introducing techniques such as accelerated inference [258] and asynchronous planning-execution [193], these approaches reduce latency and stop-and-go behaviors, resulting in smoother, more responsive, and deployment-ready navigation on real robotic platforms.

3 Context Modeling

Effective embodied navigation requires agents to continuously interpret and leverage historical observations, particularly under limited perception settings. Since a single observation often provides only partial information about the surrounding environment, accumulating and utilizing contextual information over time becomes crucial for reliable navigation. This requires the agent to understand navigation history information from multiple perspectives: perceiving spatial structure, temporal evolution, and semantic cues from observations; maintaining and retrieving relevant historical information to support long-horizon decision-making; and efficiently managing the growing amount of contextual data during navigation.

In this section, we analyze these requirements from three perspectives. We first discuss context perception, which focuses on extracting spatial and temporal information from observations. We then examine context memory, including how historical information is constructed, retrieved, and utilized during navigation. Finally, we review context compression strategies that improve the efficiency of representing and reusing contextual information. An overview and categorization of representative approaches under these three aspects are illustrated in Figure 4.

3.1 Context Perception

In navigation tasks, effective context understanding begins with the ability to perceive informative cues from observations. Under limited perception conditions, individual observations often provide incomplete or ambiguous information about the environment. As a result, agents must extract and integrate different types of perceptual signals from their observations to form a more comprehensive understanding of the surroundings. Context perception in VLN typically

involves two complementary aspects. First, spatial perception focuses on capturing structural information of the environment, enabling the agent to reason about scene layout and spatial relationships. Second, temporal perception models the sequential evolution of observations, allowing the agent to accumulate information over time and understand how the environment unfolds along the navigation trajectory. Together, these perceptual capabilities provide the foundation for building reliable contextual representations for navigation.

3.1.1 Spatial Perception. Most VLN approaches [204, 242, 243] primarily inherit visual representations from large-scale 2D vision-language pretraining, which excel at capturing semantic correlations but lack explicit spatial awareness. When applied to embodied navigation, this limitation becomes critical, as agents must reason about 3D structure and spatial relationships beyond what single-view semantics can provide. To bridge this gap, recent works exhibit a clear evolution from implicit semantic alignment toward explicit spatial modeling. The first line of research enhances cross-view consistency within 2D representations. For instance, Spatial-VLN [237] introduces a Spatial Awareness Enhancement (SPE) module that aligns panoramic observations through filtering and expert gating, improving spatial coherence without introducing explicit geometry. Building upon this, a second wave of methods injects implicit geometric priors into VLMs. Approaches such as JanusVLN [238], Efficient-VLN [238], and MG-Nav [173] leverage visual geometry models like VGGT [176] to recover 3D structural cues directly from monocular RGB inputs. These methods mark a transition from purely semantic features to geometry-aware representations, enabling better spatial reasoning without requiring additional sensors. More recent works further advance toward explicit 3D-aware modeling. For example, DyGeoVLN [111] incorporates monocular depth estimation to explicitly reconstruct 3D spatial structures, while Span-Nav [104] introduces compact spatial tokens encoding occupancy priors. These approaches reflect a broader trend: moving from implicit, feature-level spatial cues toward structured and explicitly interpretable spatial representations.

3.1.2 Temporal Perception. Temporal perception modeling in VLN has progressively expanded from simple frame selection strategies to more structured reasoning about task progression. A common and widely adopted approach is to obtain consistent timing awareness through uniform frame sampling [23, 267]. This method remains prevalent in many recent systems due to its simplicity and computational efficiency. However, such uniform sampling often fails to capture fine-grained temporal dynamics that are critical for accurate action prediction. To address this limitation, subsequent works introduce temporal prioritization mechanisms that allocate computational resources unevenly across the trajectory. For example, NavFoM [240] adopts a forgetting-curve-inspired sampling strategy, assigning higher resolution to recent frames while sparsifying distant history, thereby achieving a better balance between recency and long-term coverage. In parallel, methods such as ActiveVLN [254] and StreamVLN [204] move beyond static sampling by modeling navigation as a streaming process, incorporating dialogue-style temporal reasoning and enabling continuous action generation. Beyond improving temporal sampling, more recent research extends temporal perception toward explicit task progress modeling. While some previous approaches represent progress using simple scalar indicators [121, 180, 264], newer methods encode richer and more structured signals. For instance, M-Track [155] introduces landmark-based progress tracking, while Dual-Anchoring [205] and Progress-Think [186] integrates temporal evolution with semantic instruction grounding to enable fine-grained reasoning over task stages.

3.2 Context Memory

In VLN, long-horizon decision-making requires agents to leverage not only current observations but also accumulated experience. Due to partial observability, limited frames are insufficient to fully capture the environment, making effective context memory essential for preserving historical cues and ensuring consistent decisions. To address this,

many approaches introduce memory mechanisms that store, retrieve, and integrate past information into subsequent reasoning. Accordingly, this section reviews existing approaches to context memory from the perspectives of memory build, memory retrieval, and memory utilization.

3.2.1 Memory Representation. The design of memory representation in VLN has evolved around a fundamental trade-off between fidelity and efficiency. Some approaches primarily emphasize storing past observations, prioritizing the preservation of rich perceptual details. However, as navigation tasks extend to longer horizons, such high-fidelity representations become increasingly costly and redundant. To address this challenge, more recent methods shift toward structured abstraction and latent representations, aiming to improve memory utilization efficiency while retaining task-relevant information.

Perceptual Memory. Perceptual memory represents one of the most widely adopted paradigms in VLN, where agents retain historical sensory inputs which are typically raw image frames or encoded visual tokens as an episodic buffer. This design preserves fine-grained environmental details like textures and small objects, making it a natural choice for modern end-to-end VLN architectures. Various models adopt this memory format at different scales: from maintaining transient visual observations [145, 146, 158, 228] to constructing long-horizon episodic buffers of historical frames [8, 23, 182, 234, 235] or localized visual tokens [139, 243]. However, such static memory lacks post-hoc multi-view observability, GSMem [117] leverages 3D Gaussian Splatting (3DGS) as a persistent navigation memory, enabling agents to effectively “revisit” previously explored regions by rendering realistic views from arbitrary viewpoints, including those never physically occupied.

Symbolic Memory. Unlike perceptual memory that stores raw sensory inputs, symbolic memory abstracts observations into structured, low-dimensional representations, retaining task-relevant semantics while reducing redundancy and improving compatibility with language-based reasoning.

Many methods adopt text-driven symbolic memory for seamless integration with LLMs. HiMemVLN [119] uses a hierarchical design combining a visual topological graph based short-term memory with language-based long-term memory, while MapGPT [18] constructs a text-based topological map for flexible planning. SOL-Nav [131] fully represents history in text, mitigating the modality alignment issue between visual inputs and language models. MapNav [246] builds semantic graphs from RGB-D inputs and augment regions with textual annotations, HaltNav [90] performs room-level planning with lightweight text representations, and OpenNav [136] maintains a spatial-temporal CoT memory to track both task progress and environmental semantics. Beyond purely textual representations, several methods like OrionNav [31], OpenIN [166], FSR-VLN [263], SpatialNav [241] build memory through scene graphs that explicitly model objects and their relationships in the environment. In such approaches, historical observations are organized into graph structures where nodes correspond to objects, regions, or landmarks and edges encode spatial or semantic relations. This structured representation enables agents to maintain a coherent understanding of the scene and supports relational reasoning during navigation. By capturing both entity-level information and their interactions, scene-graph-based memories provide a compact yet expressive way to summarize past observations and facilitate downstream planning and decision making. Another prevailing strategy grounds historical semantics into spatial value maps and task states. Instead of raw pixels, these methods project language-conditioned constraints onto 2D grids to guide navigation. CA-Nav [19] builds a constraint-aware value map by continuously evaluating VLM-based similarity between current observations and sub-instruction landmarks. Extending this to complex environments, InstructNav [113] maintains multi-sourced maps that independently track actions, semantics, and historical trajectories. Furthermore, symbolic

memory excels at tracking discrete task progression; A^2 Nav [20] explicitly decomposes instructions into action-specific sub-tasks to maintain a state-based memory, while CoW [37] utilizes an open-vocabulary semantic map to dynamically switch between exploration and target-driven states. By abstracting visual noise, these structured memories drastically reduce the cognitive load and bypass the memory bloat typical of standard visual caches.

Latent Memory. Latent memory paradigm maintains a compact latent state that is continuously updated as new inputs arrive. Historical information is implicitly compressed into a fixed-size representation, allowing the model to capture temporal dependencies without incurring additional memory growth. This design enables efficient long-horizon reasoning while avoiding the overhead associated with explicit memory structures. Early navigation policies [69, 79] predominantly adopt Recurrent Neural Networks (RNNs). While effective for short trajectories, standard RNN-based memory suffers from catastrophic forgetting, leading to degraded performance as historical information gradually vanishes. To address this limitation while preserving spatial awareness, RIM [21] introduces an implicit mapping mechanism that fuses visual and geometric observations into a latent spatial grid, forming a geometry-aware memory without explicitly constructing maps. Moving beyond recurrent designs, Efficient-VLN [256] proposes a learnable key-value memory mechanism, where a fixed-size set of memory slots is recursively updated across time steps. More recently, State Space Models (SSMs) [47, 48] provide an efficient alternative. COSMO [249] leverages a Mamba-style selective scanning mechanism to model long-range dependencies with linear complexity, allowing efficient preservation of multimodal context over extended horizons.

3.2.2 Memory Retrieval and Utilization. In modern embodied navigation frameworks, memory retrieval and utilization are rarely isolated processes; rather, they are tightly coupled mechanisms where retrieved context directly parameterizes the agent’s action space. Depending on the cognitive complexity of the task, memory-driven decision-making operates across two hierarchical levels: implicit retrieval for reactive control, and explicit querying for high-level planning.

Implicit Querying. Agents require continuous context to generate smooth and stable actions, and many approaches adopt implicit retrieval mechanisms. A common starting point is to perform dense attention over accumulated perceptual histories. In perceptual memory settings, this typically corresponds to attending over the key-value (KV) cache of historical visual tokens [204, 242, 243]. However, such full-history attention introduces redundancy and computational overhead, especially in long-horizon tasks. To improve efficiency and stability, subsequent works move toward selective implicit retrieval. For example, RANGER [234] dynamically filters keyframes based on visual overlap, attending only to the most relevant historical observations. This reduces unnecessary interference and instability from redundant frames. Similarly, models like JanusVLN [238] and Efficient-VLN [256] introduce implicit memory mechanisms that query updated latent states, allowing the agent to maintain robust local geometry awareness without memory bloat.

Explicit Querying. When faced with long-horizon and ambiguous language instructions, reactive control alone is insufficient. Instead, agents must explicitly query structured symbolic memory to infer intermediate sub-goals and plan global routes. Recent works increasingly incorporate Retrieval-Augmented Generation (RAG) paradigms into navigation. For instance, MapGPT [18] queries an online linguistic topological map, leveraging the reasoning capabilities of LLMs to retrieve relevant historical nodes and generate adaptive multi-step plans. MapNav [246] explicitly injects the constructed annotated semantic graph into the LLM to enable more informed and structured navigation decisions. Scene graph-based methods [31, 166, 241, 263] adopt structured graph representations of the environment and perform explicit queries to localize target objects and reason about spatial relationships. Similarly, approaches such as CMMR-VLN [83] and MemoryVLA [152] construct multimodal experience memories indexed by panoramic observations or semantic

abstractions, enabling agents to retrieve relevant past experiences to interpret complex instructions and prune irrelevant regions. More recently, retrieval has evolved toward a predictive and anticipatory paradigm. Dual-Anchoring [205] enforces the reconstruction of dense SAM features [74] for the most recently visited landmarks, encouraging the model to preserve rich and discriminative object representations within its internal states, thereby mitigating memory decay. Memoir [221] introduces a language-conditioned world model, and retrieves past experiences conditioned on imagined future states, enabling more adaptive and human-like navigation.

3.3 Context Efficiency

As trajectory extends, accumulated multimodal context leads to a “long-context bottleneck” dominated by visual observations in both computation and memory. To enable efficient long-horizon reasoning, existing methods improve context efficiency from three aspects: temporal filtering, which selects informative frames to reduce redundancy; spatial compression, which encodes perceptual inputs into compact representations; and computational reuse, which accelerates inference by reusing intermediate features and avoiding redundant computation.

3.3.1 Reducing Temporal Redundancy. The most straightforward strategy for bounding context length is to restrict observations to a recent sliding window. Some approaches [55, 145, 146, 158] adopt this hard-truncation design, which is computationally efficient but inevitably discards long-term contextual information. To mitigate this limitation, subsequent methods [8, 23, 267] move toward *global but coarse retention* by uniformly sampling frames across the entire trajectory, preserving a global receptive field while maintaining a fixed memory budget. Building on heuristic sampling, more recent approaches shift toward adaptive keyframe extraction, aiming to retain only decision-critical observations. For instance, RANGER [234] dynamically selects keyframes by measuring the visual overlap between adjacent frames. C-Nav [235] identifies essential frames by detecting feature-space outliers, preserving observations that correspond to significant environmental changes or critical decision points. Pushing this idea further, DecoVLN [213] formulates keyframe selection as an online constrained optimization problem. It jointly considers semantic relevance, visual diversity, and temporal coverage to select a compact yet informative subset of frames. This principled formulation enables a fixed-length memory window while preserving richer semantics and broader temporal context.

3.3.2 Reducing Spatial Redundancy. Complementary to frame-level filtering, token-level compression aims to reduce redundancy within visual features themselves. Inspired by the human forgetting curve, Some approaches [131, 240, 242] primarily exploit temporal redundancy through time-aware compression that retain fine-grained tokens for recent observations while aggressively merging or pooling tokens from older frames. This design preserves short-term precision for immediate planning while controlling long-term memory growth. Building on temporal-aware compression, subsequent methods incorporate spatial awareness to further eliminate redundancy. StreamVLN [204] introduces geometry-aware token compression by leveraging 3D positional information, selectively pruning tokens corresponding to less informative regions. However, such approaches depend on accurate pose and depth signals. To relax this dependency, DyGeoVLN [111] dynamically adjusts token resolution using a learned geometric foundation model, enabling effective redundancy reduction without requiring explicit depth or pose inputs. More recently, approaches have shifted toward learnable high-ratio compression that balances efficiency and fidelity. AstraNav-Memory [139] introduces an image-centric framework with lightweight convolutional modules that aggressively compress each observation into a minimal token set. This enables a substantial expansion of the effective memory horizon while preserving essential semantic content.

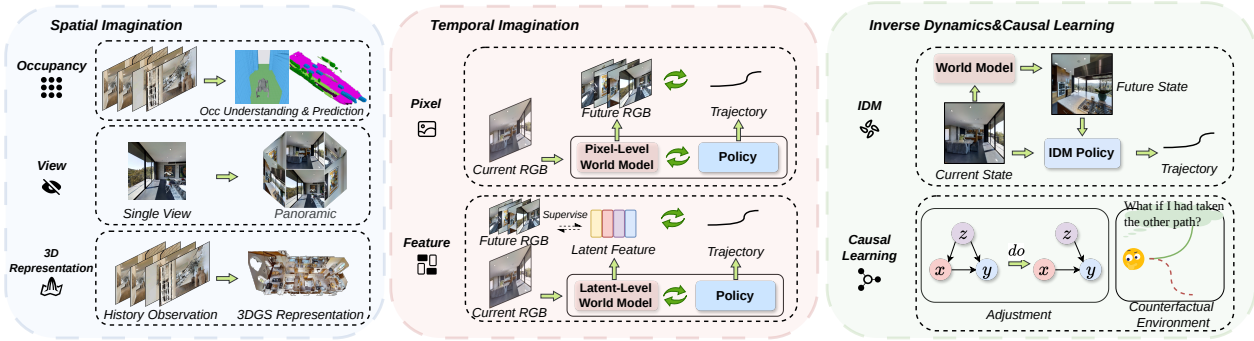


Fig. 5. Categorization of imaginative prediction approaches in VLN. We organize existing methods into three aspects: *Spatial Imagination*, which reconstructs unseen environments through semantic reasoning and 3D representations; *Temporal Imagination*, which predicts future observations and trajectories across pixel and latent; and *Inverse Dynamics and Causal Learning*, which models environment transitions and supports causal reasoning through action-state relationships.

3.3.3 Computational Reuse and Acceleration. Beyond compressing visual inputs to reduce context size, another important direction for improving context utilization efficiency is token reuse. By reusing cached representations from previous steps, this paradigm avoids redundant computation over sequential inputs, enabling faster and more responsive embodied agents.

A straightforward approach begins with direct KV-cache reuse. ActiveVLN [254] explicitly stores the full interaction history and reuses previously computed key-value states. This design suffers from increasing latency and performance degradation as the sequence length grows. To address this, methods like StreamVLN [204] adopt bounded and structured reuse, it maintain only a sliding window of recent interactions while compressing earlier context into compact history tokens using a 3D-aware pruning strategy. More advanced approaches incorporate structure-aware caching. VLN-Cache [258] selectively reuses tokens based on both viewpoint transitions and semantic changes, achieving more precise caching and improves execution efficiency with minimal performance loss.

Beyond directly operating on token caches, recent works further improve efficiency by reducing redundant computation at the architectural level. Methods like InternVLA-N1 [167], DualVLN [203], and TIC-VLA [68] decouple slow semantic reasoning from fast reactive control. In this design, high-level latent intentions can be reused across steps, enabling fast action generation. Nav-EE [59] introduces early-exit decoding to skip redundant computation, reducing inference cost without sacrificing performance.

4 Imaginative Prediction

Traditional VLN methods predominantly follow a reactive paradigm, directly mapping immediate sensory observations to navigation actions. However, such approaches are fundamentally limited by partial observability: agents lack access to critical information about occluded regions and future environmental states, both of which are essential for long-horizon planning. To overcome these limitations, agents must move beyond the constraints of their current field of view and the present moment, developing the ability to reason about the future. This capability can be broadly understood as a form of imagination, encompassing both the inference of unseen spatial structures and the prediction of how the environment evolves over time, as well as the underlying dynamics that govern such evolution.

We decompose this capability into three aspects: *spatial imagination*, which reconstructs unseen environments via semantic reasoning and 3D representations; *temporal imagination*, which models future observations in pixel or latent

space; and *inverse dynamics and causal learning*, which captures transition mechanisms and supports intervention-based reasoning. Representative methods are summarized in Figure 5.

4.1 Spatial Imagination

Navigating complex indoor environments necessitates a comprehensive understanding of scene structure. However, VLN agents are inherently limited by their ego-centric Field of View, suffering from partial observability due to occlusions and sensor limitations. To address this, current research focuses on inferring the Spatial Future, which means extrapolating unseen geometric structures and semantic layouts from partial visual cues. This capability allows the agent to construct a holistic mental map, transforming navigation from a reactive search into a planned traversal.

4.1.1 Occupancy Prediction. Occupancy prediction provides one of the most direct formulations for spatial imagination in navigation and autonomous driving, aiming to infer both the occupancy state and semantic label of each 3D location. Early approaches primarily focused on occupancy perception and static prediction [92, 194, 250, 253]. Subsequent research has evolved toward dynamic world modeling. For instance, Occworld [257] and its variants [202, 219] introduce occupancy-based world models that regress future occupancy states and agent trajectories, while enabling LLMs to reason over discretized occupancy representations. Building on this line, SparseWorld [29] constructs extended-range occupancy queries and predicts the future motion of scene elements in an ego-centric manner, removing the need for predefined spatial extents. More recent works explore representation compression for efficient reasoning. SparseOccVLA [30] unifies perception, prediction, and planning via sparse occupancy queries, while Span-Nav [104] compresses surrounding 3D occupancy into latent spatial priors that guide downstream action generation and improve navigation generalization.

4.1.2 View Completion. While occupancy-based methods enable structured reasoning over 3D space, they inherently rely on discretized representations, which can limit their ability to capture fine-grained visual details and maintain full consistency with raw sensory observations. To address this gap, subsequent research has shifted toward view-based spatial imagination, enabling more realistic and perceptually grounded predictions. Some works such as PanoGen[85] and PanoGen++[187] leverage generative models to synthesize panoramic semantic observations for unseen regions, improving the coverage and continuity of the perceived environment. This marks a transition from deterministic reconstruction to probabilistic, generative scene completion. Extending this paradigm, more advanced methods move toward joint spatial imagination and policy learning within unified frameworks. MapDream [94] integrates spatial representation learning with decision-making by autoregressively generating semantic bird’s-eye-view maps conditioned on textual instructions and egocentric observations. In parallel, another line of work focuses on efficient latent-space imagination. MonoDream [185] trains a vision-language model to predict panoramic RGB and depth representations from monocular inputs in a latent space, striking a balance between realism and efficiency in future view synthesis.

4.1.3 3D Representation. Furthermore, the representation of the completed world is shifting towards explicit 3D representations to improve prediction quality. As highlighted in the recent survey on 3D Gaussian Splatting [54], 3DGS offers a continuous and differentiable scene representation. Unlike Neural Radiance Fields (NeRFs), 3DGS supports real-time rendering, enabling the agent to perform *Novel View Synthesis* for unvisited areas. This effectively allows the agent to “imagine” the visual appearance of a destination before arrival, bridging the gap between spatial topology and visual perception. Building on this paradigm, MonoVLN [115], UnitedVLN [28], ThinkMatter [27], and GSMem [117] infer representations of missing regions using 3DGS, allowing agents to reason about unseen space through structured

3D primitives. This explicit modeling further bridges spatial topology and perceptual imagination, strengthening anticipatory navigation.

4.2 Temporal Imagination

Effective planning requires agents to reason about the temporal future, i.e., to anticipate the consequences of potential actions before execution. Recent advances have shifted to generative imagination, enabling agents to simulate future states at multiple levels of abstraction, including pixel-level visual observations and feature-level representations.

4.2.1 Pixel Level. Recent studies have shown that equipping agents with visual imagination, defined as the ability to anticipate future observations prior to execution, can significantly enhance decision-making [76, 133]. At the most fine-grained level, this capability is instantiated through pixel-level prediction, where agents leverage video generation techniques to perform look-ahead in the raw observation space. Conditioned on the current observation and a candidate action sequence, these models synthesize future RGB frames, effectively acting as neural simulators of the environment.

NWM [9] formulates visual navigation as a controllable video generation problem conditioned on past frames and actions. These methods establish the foundation of action-conditioned visual prediction. Subsequent works introduce stronger generative priors and planning-oriented extensions. VISTA [67] leverages diffusion models to perform action-conditioned imagination guided by local observations and language instructions. VISTAv2 [66] augments this with an imagined value prediction head, integrating generated future views into an online value map. In parallel, NavDreamer [65] adopts a generate-then-select paradigm, generating multiple candidate video clips conditioned on the given instructions, then employs a VLM to filter them based on instruction alignment and environmental constraints. These approaches mark a shift toward decision-aware imagination, where generated visuals are explicitly leveraged for downstream planning. More recent work moves beyond modular-style designs toward tightly coupled world modeling and policy learning. AstraNav-World [60] unifies visual imagination and action prediction within a single framework. It jointly models future visual frames and multi-step action sequences conditioned on the current state and linguistic instructions.

Despite their effectiveness, pixel-level imagination methods suffer from high computational cost and sensitivity to stochastic noise due to dense frame generation. To mitigate these issues, recent efforts explore efficiency-oriented abstractions of visual imagination. SparseVideoNav [239] reduces both training and inference overhead by adopting sparse video generation, enabling longer-horizon prediction with fewer resources. Meanwhile, DreamNav [195] equips its world model with a Narration Expert, which abstracts dense visual sequences into concise semantic narratives, thereby improving computational efficiency and interpretability while reducing redundant information. Nevertheless, such methods fundamentally remain within the paradigm of pixel-space imagination.

4.2.2 Feature Level. Different from pixel-level imagination, recent works have increasingly shifted toward latent-space visual imagination, where future observations are predicted in compact feature representations rather than raw pixel space. This transition improves efficiency while retaining semantically meaningful information for downstream decision-making. MonoDream [185] predicts future panoramic RGB and depth features using only monocular RGB input, performing future imagination directly in a learned feature space rather than explicit 3D reconstruction. This latent prediction mechanism improves efficiency while preserving semantic coherence in anticipated views. LatentPilot [52] employs propagated and compact visual latents that encode imagined near-future observations, enabling the learning of motion-conditioned visual dynamics in the latent space. NavMorph [229] uses a world-aware navigator to infer environmental dynamics and construct latent states. A foresight planner then generates navigation controls while

decoding these states into predictive visual embeddings, providing scene-aware information for decision-making. NavForesee [101] employs a set of learnable Dream Queries to predict a compact collection of high-level representations, including depth, DINOv2 [126], and SAM [74] features. PROSPECT [35] similarly uses a set of learnable streaming queries to predict the 2D and 3D latent features for the next time step. In contrast, FantasyVLN [268] adopts an autoregressive strategy to generate future frames within the VAR [168] latent space, achieving a higher compression ratio while preserving reconstruction fidelity.

4.3 Inverse Dynamics Modeling and Causality

Generative world models can be seen as forward dynamics models, predicting future states based on current observations. In contrast, inverse dynamics models aim to infer the actions required to achieve a desired state transition, offering a deeper understanding of environmental causality and underlying physical principles. In embodied visual-language navigation, inverse dynamics provides a computationally efficient way to embed knowledge of actions and the world directly into the policy space, enforcing a deterministic mapping between visual changes and agent motion and thereby reducing error accumulation over long trajectories. This section focuses on applications of inverse dynamics in VLN and explores how causal reasoning can help distinguish genuine navigational dependencies from spurious correlations.

4.3.1 Inverse Dynamics. Understanding the future also requires understanding the past. Inverse dynamics models $P(a_t | s_t, s_{t+1})$ infer the action taken to cause a state change. In the context of VLN, this supports *Imitation Learning* by helping the agent understand the intent behind expert demonstrations. Inverse dynamics pretraining has been shown to learn robust and effective visual representations while maintaining training efficiency and stability [10]. Building on this idea, LAPA [230] and UniVLA [12] leverage inverse dynamics objectives to acquire implicit robotic skills from large-scale, unsupervised video data. Seer [169] employs an end-to-end predictive inverse dynamics model to achieve closed-loop integration of imagination and action. NAVIDA [267] and SpaAct [89] introduce inverse dynamics supervision to inject language-independent, action-grounded visual dynamics into the model, thereby enhancing its understanding of action semantics. SparseVideoNav [239], LatentPilot [52], NavDreamer [65] and ImagiNav [16] further integrate inverse dynamics with visual imagination, they first generate sparse future video frames and then employ an inverse dynamics formulation to infer the corresponding actions by reasoning over temporal visual consistency between preceding and predicted frames.

4.3.2 Causal Learning. In practice, agents often overfit to dataset biases and spurious cues—for example, associating a specific carpet pattern with a “move forward” action. To achieve true generalization, agents must move beyond statistical correlations toward causal reasoning. Some works focus on explicit causal intervention through structural modeling. GOAT [179] introduces a unified structural causal model for VLN, explicitly modeling both observable and latent confounders. By applying the do-operator, it simulates interventions on feature distributions to block spurious information flow, and addresses confounding through backdoor and frontdoor adjustment modules. Similarly, CausalVLN [178] adopts iterative backdoor-based representation learning, enabling adaptive interventions during training. Beyond intervention, counterfactual reasoning offers another powerful paradigm, allowing agents to ask: “Would a different action lead to better outcomes?” Methods such as CVLN-Think [107], CF-VLN [174], and CAST [43] train agents in counterfactual environments (e.g., altered layouts or instructions), encouraging them to learn invariant causal relationships between language and navigation. This significantly improves robustness in unseen environments.

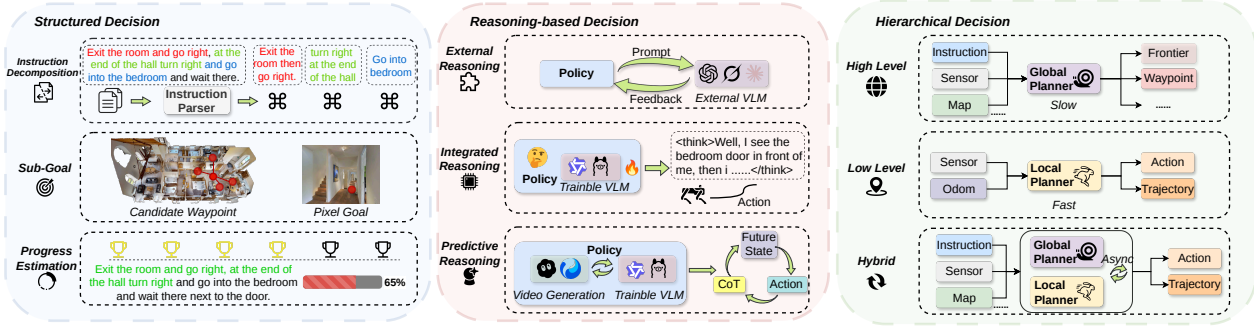


Fig. 6. Categorization of decision planning approaches in VLN. We organize existing methods into three aspects: *Structured Decision*, including sub-goals, instruction decomposition, and navigation progress perception; *Reasoning-based Decision*, which introduces intermediate reasoning trajectories to bridge perception, instruction understanding and action; and *Hierarchical Decision*, which coordinates high-level planning and low-level control across different temporal and semantic scales.

5 Decision Planning

Visual-Language Navigation is a sequential decision-making problem under partial observability, requiring agents to transform incomplete observations and high-level instructions into temporally consistent actions. Performance depends not only on perception, but on how decisions are structured and coordinated over time. Most VLN methods [204, 238, 242, 243, 254, 267] adopt reactive mappings from observations and instructions to actions, implicitly encoding decision logic. While effective in simple settings, such designs struggle in continuous and monocular scenarios due to uniform reasoning and error accumulation. This motivates a shift toward explicit decision organization, where high-level planning, sub-goal reasoning, and low-level control operate at different scales.

This section reviews decision in three aspects: *structured decision*, including instruction decomposition, sub-goals and progress estimation; *reasoning-based decision*, including external reasoner, integrated reasoner and predicted reasoning model; and *hierarchical decision architectures* that coordinate high-level planning with reactive control. Representative methods are summarized in Figure 6.

5.1 Structured Decision

Effective navigation requires more than instantaneous action decision, it depends on how decisions are organized over time. Here we mainly focus on three key elements of structured decision modeling: instruction decomposition, sub-goal formulation, and navigation progress estimation.

5.1.1 Instruction Decomposition. One line of works leverage LLMs to explicitly decompose instructions into sub-instructions. DILLM-VLN [177] leverages lightweight open-source LLMs to decompose complex navigation instructions into a sequence of concise and interpretable sub-instructions. The agent is subsequently trained via reinforcement learning to complete these sub-instructions in a sequential manner. Spatial-VLN [237] introduces a multi-expert scheme, decomposing instructions into parallel waypoint and region sequences for structured and complementary reasoning over fine-grained objects and broader spatial regions. Another line of works formulate instruction decomposition as an optimization problem. CA-Nav [19] describes navigation as a sequential, constraint-aware process of completing sub-instructions. It defines constraints from object, positional, and directional perspectives, enabling structured transitions between sub-instructions during execution when these constraints are satisfied. GC-VLN [231] decomposes language instructions into a Directed Acyclic Graph (DAG) and queries the constraint library to instantiate graph-level constraints.

5.1.2 Sub-Goal Formulation. Early approaches such as SmartWay [153] and GridMM [199] typically adopt a two-stage framework, where a high-level module first generates a set of candidate waypoints, after which the agent selects the most suitable sub-target according to the navigation instructions [57]. However, this design depends heavily on waypoint quality and often lacks rich semantic grounding [150]. To address these limitations, subsequent works enhance sub-goal generation by incorporating external knowledge and LLM-based reasoning. LGK [225] retrieves commonsense knowledge via CLIP alignment and refines it with landmark cues. The plan agent in MerNav [135] generates concise sub-objectives from multimodal context, which serve as an intermediate layer between coarse-grained direction filtering and fine-grained action selection. Building upon this, another line of work focuses on grounding sub-goal generation in explicit visual and spatial cues. AO-Planner [17] introduces visual affordances into the navigation pipeline. It employs SAM to segment visible ground regions and generate Visual Affordance Prompts that indicate navigable areas. Based on these prompts, an LLM selects feasible waypoints and plans paths in pixel space. DualVLN [203] formulates high-level planning as locating the farthest reachable pixel target. A VLM predicts the 2D coordinates of the next waypoint in the image, improving the interpretability of planning. More recent approaches aim to formalize the construction of sub-goals from a principled mathematical perspective. HSAN [36] constructs a dynamic hierarchical semantic scene graph to represent the environment. It leverages a VLM to build multi-level scene representations and formulates navigation target selection as an optimal transport problem. The Sinkhorn algorithm is employed to compute a principled balance between the semantic relevance and physical reachability of candidate sub-targets, providing a mathematically grounded mechanism for robust sub-goal selection.

5.1.3 Navigation Progress Estimation. During navigation, an agent’s physical movement does not necessarily translate into meaningful progress toward completing the instruction. Consequently, navigation policies must explicitly estimate progress to regulate the decision process, allowing the agent to adjust subsequent decisions to remain aligned with the intended goal. Early efforts primarily model progress as scalar signals for auxiliary supervision [121, 180, 264]. However, these scalar signals lack explicit semantic information [120]. To overcome this limitation, subsequent work incorporates structured and semantically grounded progress signals. M-Track [155] proposes a milestone builder that annotates instructions with navigation and interaction milestones to be completed sequentially, along with a milestone checker that evaluates whether the current milestone has been achieved. Spatial-VLN [237] introduces two dedicated progress estimation experts to monitor execution status based on instruction decomposition. These modules leverage hierarchical observations and task decomposition to predict the completion progress of the current waypoint. Similarly, CA-Nav [19] tracks navigation progress through explicit sub-instruction switching derived from instruction decomposition. More recent work moves toward semantic progress modeling, where completed portions are treated as prefixes of the full instruction. Dual-Anchoring [205] enforces the model to explicitly articulate its navigation progress state before predicting actions, while Progress-Think [186] introduces a three-stage framework for self-supervised semantic progress estimation.

5.2 Reasoning-based Decision

Chain-of-thought (CoT) reasoning [192, 201] intermediate reasoning steps, enabling models to decompose complex tasks into a sequence of simpler decisions. In VLN, CoT provides a bridge between perception, instruction understanding, and action selection, helping agents reason over long-horizon tasks under partial observability. Unlike direct observation-to-action mappings, CoT-based methods explicitly generate intermediate intentions or plans, improving interpretability and reducing error accumulation. Existing reasoning-based VLN methods mainly differ in where reasoning is performed and

how tightly it is coupled with navigation. Early approaches use LLMs as external reasoners for zero-shot planning, while later end-to-end methods internalize reasoning within the policy itself. More recent work further integrates multimodal imagination, enabling reasoning over predicted future observations or latent world states. Accordingly, we categorize reasoning-based VLN methods into three paradigms: *LLMs as External Reasoners*, *Integrated Perception-Reasoning-Action Models*, and *Predictive Reasoning Models*. Together, these paradigms reflect a shift from tool-based planning toward intrinsic and cognitively inspired navigation reasoning.

5.2.1 LLM as External Reasoner. Using off-the-shelf LLMs and VLMs as external reasoners provides a lightweight solution for navigation planning. In this paradigm, explicit chain-of-thought prompting guides decision-making, while the navigation policy remains loosely coupled with the reasoning process and is free from reasoning training.

One line of works convert navigation into textual reasoning. InstrucNav [113] constructs a CoT-like Dynamic Chain of Navigation to structure the navigation process. OrionNav [31] queries GPT-4 Turbo to generate navigation plans together with explicit reasoning processes, and OpenNav [136] decomposes navigation into instruction understanding, progress estimation, and decision-making with CoT prompting.

Another line incorporates visual information into reasoning. VLMNav [44] formulates navigation as visual question answering over candidate actions, ImagineNav++ [189] reasons over synthesized future views and then select the final action, and SpatialGPT [70] introduces a Synchronize-Align-Backtrack (SAB) reasoning chain, achieving a balanced exploration exploitation trade-off through explicit reasoning and backtracking.

The third line of works are inspired by human navigation behavior and performs reasoning over top-down spatial representations. TopV-Nav [259] and ReasonNavi [5] employ multimodal LLMs on semantic maps, while GeoNav [214] combines global cognitive maps and local scene maps with multimodal CoT reasoning for interpretable navigation.

5.2.2 Integrated Perception-Reasoning-Action Model. While external LLM/VLM reasoners offer flexibility, they rely heavily on carefully crafted prompts and loosely couple perception, reasoning, and action. This has motivated end-to-end VLN models that internalize chain-of-thought (CoT) reasoning within trainable perception-reasoning-action pipelines.

One line of works integrates CoT through supervised learning. NavCoT [97] converts observations into text and trains an LLM to jointly perform world modeling and reasoning, while D3D-VLP [197] introduces a Dynamic 3D CoT framework for unified planning, grounding, navigation, and question answering.

Subsequent methods further enhance reasoning with reinforcement learning. Nav-R1 [105] and MobileVLA-R1 [63] combine CoT supervision with GRPO-based RL to encourage reasoning before action. OctoNav [38] further incorporates an online reinforcement learning phase, enabling agents to learn proactively through trial-and-error interaction with the environment. CompassNav [88] proposes a Gap-Aware Hybrid Reward Function to handle ambiguity in low-deterministic scenarios while providing decisive guidance when navigation outcomes are more certain.

However, excessive reliance on explicit reasoning can introduce significant computational overhead during deployment. To address this issue, recent work explores adaptive and implicit reasoning. HiRO-Nav [255], AdaNav [33], VlingNav [183], and Hydra-Nav [196] introduce Adaptive CoT, enabling agents to switch between fast execution and more deliberate reasoning, while AgentVLN [212] combines semantic reasoning with skill invocation through query-driven CoT. The other direction aims to internalize reasoning within the model to reduce reliance on explicit chain-of-thought during inference. Aux-Think [184] uses CoT as auxiliary supervision while adopting no-think inference, and Abot-N0 [24] introduces an inference task encoder that activates chain-of-thought reasoning within the LLM by injecting structured task descriptions into intermediate blocks.

5.2.3 Predictive Reasoning Model. ImagineNav++ [189] formulates navigation as imagination-driven viewpoint selection: the agent predicts candidate waypoints, synthesizes future views, and uses a VLM to reason over imagined observations for waypoint selection. VISTA [67] similarly generates prospective target views with Stable Diffusion and combines perceptual alignment with LLM-based reasoning, enabling decisions conditioned on anticipated future states. NavForesee [101] integrates hierarchical language planning with dual-view predictive look-ahead to unify multimodal planning and prediction. It organizes navigation into completed sub-instructions, the current sub-goal, and future semantic actions to form a high-level plan. Moreover, it functions as a world model, imagining future high-level semantic features across both short and long-term horizons. FantasyVLN [268] further integrates Text-CoT, Visual-CoT, and Multimodal-CoT within a unified latent-space framework [168], improving reasoning consistency and efficiency through cross-modal alignment and latent visual reasoning.

5.3 Hierarchical Decision

Effective embodied navigation decision depends not only on reasoning ability but also on how planning and control are organized within the system architecture. Existing methods typically emphasize different levels of this hierarchy. High-level systems focus on semantic reasoning and long-horizon planning but often suffer from high latency, while low-level systems provide efficient and responsive control but lack global reasoning and language understanding. Recent work increasingly explores hybrid architectures that combine high-level planning with low-level reactive control, enabling both strategic reasoning and robust execution. This section reviews these navigation architectures and their multi-level decision organization.

5.3.1 High-Level System. High-level navigation systems provide global guidance for navigation tasks, formulating strategies such as selecting key waypoints or determining directions of exploration rather than directly generating paths or low-level actions [31, 166, 259]. Operating primarily on semantic or map-level representations, they maintain consistency between local behavior and long-horizon goals. VLFM [232] combines occupancy map and VLM-based value maps to select informative frontier waypoints, while InstrucNav [113] integrates multiple value maps for waypoint planning. OpenFrontier [127] uses visual boundaries as semantic anchors for target-aware planning and exploration. MerNav [135] identifies traversable regions from depth maps and then prompting action agent to evaluate these candidates to determine the optimal location.

5.3.2 Low-Level System. Low-level navigation systems are typically vision-driven point-goal local planners that directly map sensory inputs to planned paths or control actions [56, 114, 145, 146]. Early methods mainly rely on imitation learning which often requires large amounts of supervised data [114, 143, 146]. While reinforcement learning offers a trial-and-error optimization framework, its low sample efficiency and sparse reward signals make training computationally expensive [151, 222]. Recent work explores more scalable training paradigms. iPlanner [226] adopts an imperative learning paradigm and formulates path planning as offline bi-level optimization for faster convergence. However, it relies on real-world data to bridge the sim-to-real gap. ViPlanner [142] combines semantic and geometric inputs to achieve sim-to-real transfer using simulation-only training. NavDP [14] further scales simulation data and exploits privileged information for robust trajectory generation and cross-ontology generalization. Despite improved efficiency and generalization, these methods often depend on external localization. To address this, LogoPlanner [132] incorporates implicit state estimation through visual geometry reconstruction, enabling more reliable localization and obstacle avoidance.

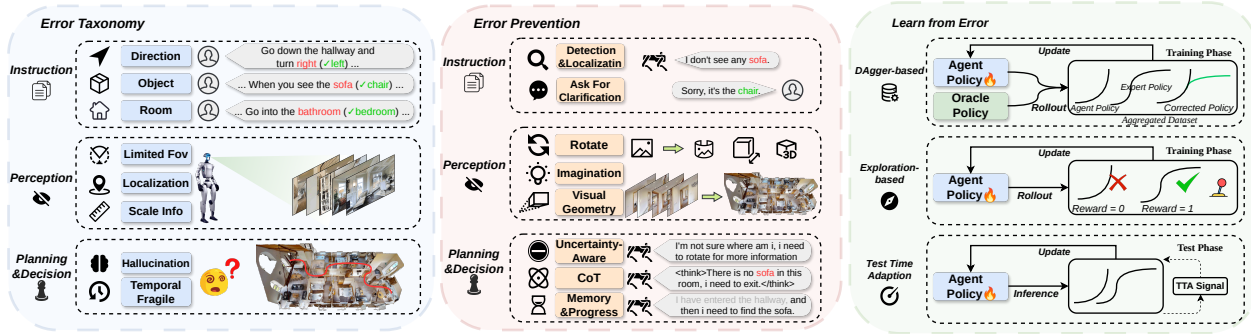


Fig. 7. Categorization of error-aware and reflection-based approaches in VLN. We organize existing methods into three aspects: *Error Source Analysis*, which identifies failures arising from instruction errors, perception errors, planning and decision errors; *Error Prevention*, which anticipates risks through methods like uncertainty-aware decision-making and conservative planning; and *Learning from Errors*, which leverages failures as supervisory signals for continuous policy improvement.

5.3.3 Hybrid Architecture. High-level navigation systems provide strong semantic reasoning and long-horizon planning but often suffer from high latency and limited motion sensitivity, while low-level systems generate smooth and responsive motions yet lack global reasoning, and often depend on externally specified targets. To bridge this gap, recent work explores hybrid architectures that combine semantic planning with low-level control.

A common strategy is hierarchical decomposition, where high-level modules generate sub-targets and low-level planners execute feasible motions. LaViRA [32] organizes navigation into language, visual, and robot action levels. Building on this hierarchical structure, dual-system architectures explicitly separate slow reasoning from fast control. DualVLN [203] and InternVLA-N1 [167] use large VLMs to perform semantic planning and infer intermediate pixel-level targets, and lightweight local planners to generate continuous trajectory, while OmniNav [223] couples slow CoT reasoning with fast waypoint control through a central memory module implemented with a KV cache. Recent methods further improve coordination between reasoning and execution. TIC-VLA [68] introduces delayed semantic control interface for asynchronous reasoning, and Nav-R1 [105] reuses representations from the slow system to enable semantic consistency in planning and efficient low-latency control. In parallel, modular approaches such as SkillNav [122] and AgentVLN [212] decompose navigation into reusable skills, using VLMs to dynamically route between perception and planning modules for improved generalization.

6 Reflection from Error

Monocular and mapless VLN poses fundamental challenges for embodied agents, including perceptual aliasing, spatial ambiguity, and long-horizon error accumulation. In end-to-end systems, tightly coupled perception, reasoning, and control further amplify cascading failures. Robust VLN therefore requires not only performance improvements but also systematic understanding of how errors arise, interact, and evolve.

This section reviews error handling mechanisms in LLM-based end-to-end VLN under sensor-constrained settings. We first analyze error sources from instruction understanding, perception, reasoning, and planning. We then discuss error prevention strategies, such as uncertainty-aware decision-making and conservative planning, followed by methods that learn from failures to improve policies through self-correction and reflection. Representative approaches are summarized in Figure 7.

6.1 Sources and Taxonomy of Errors in VLN

6.1.1 Instruction Errors. Language instructions serve as the primary supervisory signal in visual language navigation, yet most benchmarks assume they are accurate and unambiguous. In practice, human-generated instructions are often noisy due to limited perception, imprecise spatial memory, and cognitive bias. Prior work [161, 164] explicitly models such instruction-level errors and shows their significant negative impact, revealing the fragility of VLN systems under imperfect linguistic supervision.

Instruction errors typically arise in three forms. *Direction errors* are particularly detrimental, as monocular agents rely on relative terms (e.g., left, right, forward) without a global reference frame; incorrect directions can cause large deviations, with reported success rate drops exceeding 18.6% [164], and are hard to recover from once key visual cues leave the field of view. *Object and landmark errors* occur when annotators misidentify or misdescribe entities (e.g., category or attributes such as color and size), especially for commonly co-occurring objects, leading to failures in visual grounding; this issue is further exacerbated in RGB-only settings by illumination changes, viewpoint variation, and occlusion [181]. *Room and region errors* involve incorrect high-level spatial descriptions, where semantically or visually similar areas are confused; without global spatial awareness, monocular agents struggle to detect such inconsistencies, resulting in compounded errors over long-horizon navigation.

6.1.2 Perceptual Errors. Most prior VLN systems [3, 22, 97, 260] assume access to panoramic observations, where navigation reduces to selecting the best-matching viewpoint within a predefined graph. This setting relies on strong assumptions, including known environment topology, perfect localization, and omnidirectional perception. In contrast, real-world agents typically operate with a single forward-facing camera, fundamentally constraining perception and introducing additional errors. *Limited field-of-view errors* arise because monocular observations are narrow and egocentric, leaving large portions of the environment unobserved, especially in lateral and rear directions. Consequently, the agent’s spatial understanding is fragmented and must be constructed via temporal integration across frames [115]. This dependence on memory accumulation increases susceptibility to perceptual aliasing and makes errors difficult to detect or recover from once critical visual cues are missed. Further perceptual errors like *depth and scale ambiguity* and *localization and motion estimation errors*, can be similarly categorized under this constrained sensing paradigm.

6.1.3 Planning and Decision Errors. As an increasing number of VLN methods adopt LLMs and VLMs as the core framework for planning and decision-making, model-level planning and decision errors have become more salient. Incomplete or ambiguous observations can disrupt the integration of memory, language, and perception, resulting in inconsistent internal beliefs. *Prior hallucination* arises when strong semantic priors in pre-trained models induce hallucinated reasoning and over-reliance on learned correlations instead of real-time visual evidence [6, 61, 62, 170, 248]. Meanwhile, *temporal fragility* emerges in long-horizon tasks, where limited context capacity, fragmented memory, and weak progress awareness progressively degrade reasoning consistency and decision quality [157, 186, 209, 264]. Together, these factors lead to persistent and compounding errors in VLN.

6.2 Error Prevention

6.2.1 Prevent Instruction Errors. Instruction-level errors mainly stem from ambiguous and complex human instructions and mismatches between referenced landmarks and the actual environment. A common strategy for preventing such errors is to clarify instructions before or during execution. IEDL [164] introduces an instruction error detector and locator to detect inconsistent instructions and localizes erroneous segments. However, it operates only after navigation

is completed. To enable online correction, I2EDL [163] extends this framework with interactive clarification, allowing the agent to detect potential errors during navigation and query users for corrective feedback. Similarly, AIUTA [165] autonomously requests more detailed target descriptions when coarse instructions are insufficient. Beyond explicit interaction, another direction investigates uncertainty-aware and minimally interactive strategies. SwPC [220] adopts conformal prediction to calibrate uncertainty over open-vocabulary outputs, thereby reducing the need for human intervention. Conformal prediction provides distribution-free uncertainty guarantees with minimal assumptions about the underlying model.

6.2.2 Prevent Perception Errors. As mentioned earlier, errors at the perception level mainly stem from limited perceptual field of view, lack of depth and other scale information, and the absence of precise positional information. One intuitive solution is to force the agent to rotate 360 degrees in place at each time step to achieve panoramic perception [232]. While this method is easily implemented, it significantly reduces the agent’s execution efficiency, requiring the agent to perform more than twice the number of actions [115]. To overcome these limitations and improve efficiency, subsequent approaches shift toward perception completion from limited observations. MonoDream [185] trains its model to predict latent features of panoramic RGB and depth perception in the current and future steps based solely on monocular RGB input. MonoVLN [115] uses 3DGS to leverage contextual feature relationships to infer representations of missing regions rendered from the feature field. More recent methods like MG-Nav [173], JanusVLN [238], DyGeoVLN [111] and Efficient-VLN [256] embed visual geometry models into the VLN framework to extract geometric information containing depth and position from the raw RGB frame sequence, enabling the model to understand spatial structure during navigation.

6.2.3 Prevent Planning and Decision Errors. Preventing planning and decision-making errors fundamentally hinges on mitigating agent illusions. Achieving this goal typically requires agents to maintain long-term representations of the environment while simultaneously monitoring their own cognitive and navigational states in real time. By identifying high-risk decisions early, agents can adopt more conservative action strategies or proactively seek additional information before committing to potentially irreversible moves. AdaNav [33] and HiRO-Nav [255] leverage action entropy to trigger additional CoT reasoning under uncertain conditions. MonoVLN [115] further introduces an uncertainty-aware active perception policy that combines directional novelty with entropy estimation. When uncertainty is high, the agent suppresses forward movement and instead performs exploratory “check” actions to gather additional observations before committing to potentially irreversible decisions.

Beyond uncertainty-aware control, recent works improve planning robustness through explicit reasoning and memory modeling. Several methods [38, 63, 105, 183, 184] introduce chain-of-thought reasoning to enhance task decomposition, language–perception alignment, and decision consistency. EvolveNav [98] further adopts self-reflective reasoning by constructing positive and negative reasoning pairs from model-generated CoTs, improving reasoning robustness through iterative training. Other approaches focus on long-horizon memory and progress awareness: MGDM [157] integrates short- and long-term memory for dynamic environments, SeqWalker [51] combines instruction segmentation with scene graphs for sequential navigation, and Progress-Think [186] introduces semantic progress awareness for more consistent decision-making.

6.3 Learn From Error

In continuous environments with constrained sensors, simple teacher-forcing behavior cloning (BC) often generalizes poorly to unseen scenarios. This limitation stems from the fact that training data are typically generated by an oracle

and consist almost exclusively of optimal or near-optimal trajectories. During deployment, however, agents inevitably accumulate small execution errors, gradually drifting into state distributions that were never encountered in the expert demonstrations, which is a phenomenon commonly referred to as covariate shift [106, 183, 254]. In these unfamiliar states, the agent lacks appropriate supervision, causing errors to rapidly compound and leading to navigation failure [218]. Consequently, learning from failure and off-distribution states is crucial for improving navigation robustness.

6.3.1 Dagger-based. DAgger [141] provides a classic solution to covariate shift by querying an oracle for corrective actions under the agent’s own state distribution and iteratively aggregating these samples into training. Building on this framework, Efficient-VLN [256] introduces a dynamic hybrid policy that progressively biases policy selection toward the oracle as navigation proceeds, while DifNav [150] incorporates DAgger-based online policy training and expert trajectory augmentation to improve robustness in erroneous states. Beyond oracle-assisted correction, CorrectNav [236] and LatentPilot [52] adopt data flywheel mechanisms that automatically collect failed trajectories and generate self-correcting supervision from action, perception, or privileged target information. Through iterative retraining, these methods progressively enhance the agent’s ability to recover from errors.

6.3.2 Exploration-based. In contrast to imitation-based approaches, reinforcement learning (RL) provides a more proactive paradigm for error exploration and environment interaction. Most recent methods [84, 149, 183, 254] are built upon the Group Relative Policy Optimization (GRPO) [147] framework and its variants. ActiveVLN [254] and NavGRPO [87] encourage active exploration and diverse trajectory generation through outcome-based rewards, though such optimization often suffers from sparse reward supervision. To address this issue, SeeNav-Agent [200] and MobileVLA-R1 [63] introduce step-level rewards that provide dense feedback during navigation. VLingNav [183] further incorporates expert supervision into agent–environment interaction to balance exploration and exploitation, while BudVLN [53] improves exploration efficiency through an online retrospective framework that dynamically combines RL and supervised learning. More recent approaches explore extracting informative supervision from imperfect trajectories. SACA [84] extracts supervision from failed trajectories using a Perception-Grounded Step-Aware auditor, which identifies valid trajectory prefixes and divergence points to provide dense learning signals instead of discarding imperfect samples.

6.3.3 Test-time Adaption. Test-time adaption (TTA) [95, 211] has emerged as an effective strategy for mitigating navigation errors under distribution shift. Unlike offline learning methods that rely on expert data or retraining, TTA adapts the agent online using weak feedback signals during deployment, making it particularly appealing for monocular VLN in continuous environments. FEEDTTA [73] adopts an online RL framework that updates the policy using binary success/failure feedback during testing, enabling correction of behavioral biases under domain shift. FSTTA [40] formulates adaptation as an uncertainty minimization problem and prevent instability caused by aggressive updates through cumulative gradient and parameter analysis. Building on this idea, ATENA [75] proposes a hybrid entropy optimization strategy that minimizes entropy for successful trajectories to reinforce reliable actions, while maximizing entropy for failed trajectories to discourage overconfident incorrect decisions and encourage selectively consolidate correct behaviors while maintaining exploration in uncertain or erroneous states.

7 Future Directions

Despite the rapid progress of LLM and VLM-based visual language navigation in continuous environments, existing methods remain constrained by persistent challenges in perception, reasoning, decision-making, and real-world deployment, and are still far from achieving truly robust, general, and deployable embodied intelligence. In this section,

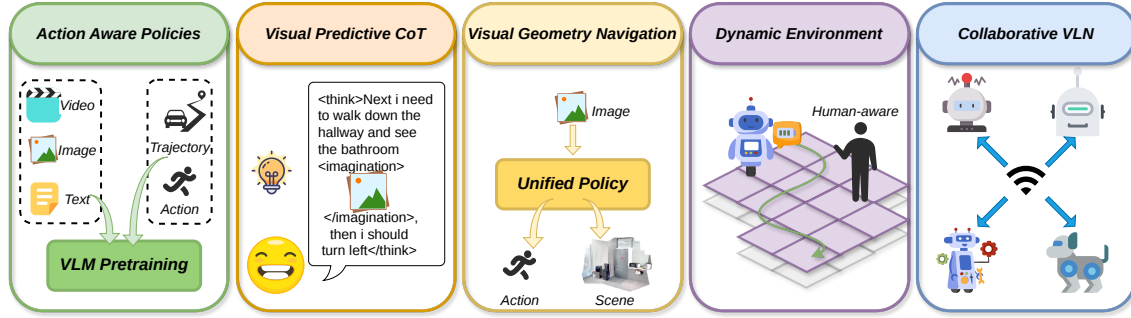


Fig. 8. Overview of future research directions for LLM/VLM-based VLN, including action-aware policy learning through embodied training, visual predictive chain-of-thought reasoning, unified visual–geometry navigation systems with persistent spatial representations, adaptive navigation in dynamic and socially interactive environments, and collaborative multi-agent frameworks with shared perception and coordinated decision-making.

we discuss the key limitations of current VLN approaches and highlight several promising research directions that may help overcome these challenges, as can be seen in Figure 8. Our goal is to outline pathways toward more robust, generalizable, and practically applicable LLM/VLM-driven VLN systems.

7.1 Towards Action-Aware Policies

Most existing VLN methods [204, 242, 243] rely on task-specific post-training or fine-tuning of large LLMs and VLMs. While this paradigm substantially enhances instruction comprehension and high-level semantic reasoning, it remains fundamentally constrained by the pre-training regimes of current VLMs. In particular, most VLMs [7, 82, 112] are predominantly trained on static image–text or video–text pairs and web-scale multimodal data, with little to no exposure to embodied interaction, action execution, or long-horizon navigation trajectories. As a result, the “actions” employed in downstream VLN tasks—typically represented as discrete labels or linguistic commands—are weakly grounded in physical dynamics, egocentric motion, and temporal causality.

A promising direction is to incorporate navigation and interaction-centric objectives directly into the pre-training or intermediate training stages of VLMs. By exposing models to large-scale trajectory data and action-conditioned state transitions, such training can enable VLMs to jointly associate visual observations, linguistic descriptions, and the environmental changes induced by actions. Potential approaches include world-model-based objectives, action-conditioned future observation prediction, and other forms of self-supervised embodied learning.

7.2 Towards Visual Predictive CoT

Recent VLN research has increasingly explored chain-of-thought (CoT) reasoning [183, 184] and world models [185, 229] as complementary paradigms for long-horizon decision-making [96]. CoT-based methods improve interpretability by explicitly verbalizing intermediate reasoning steps, but remain fundamentally text-centric and often struggle to faithfully represent spatial layouts, geometric constraints, and fine-grained visual relationships in partially observable environments. In contrast, world models predict future observations, spatial states, or latent representations conditioned on actions, enabling anticipation beyond reactive control. However, existing approaches are largely visual-centric and rarely integrated into high-level reasoning, limiting their semantic interpretability and task awareness. This separation fragments the agent’s cognition: world models predict future states without semantic intent, while reasoning modules plan actions without simulating their physical consequences.

A promising direction is to tightly couple CoT reasoning with explicit world or forward dynamics models [67, 268], enabling visual- and geometry-grounded chains of thought. In this paradigm, predicted future observations and latent state transitions become intermediate cognitive states within reasoning and planning, allowing agents to reason through imagined futures, evaluate alternative actions before execution, and align semantic goals with anticipated environmental outcomes. Such integration offers a principled path toward more reliable, interpretable, and generalizable VLN systems under sensor-constrained and long-horizon settings.

7.3 Towards Unified Visual Geometry Navigation Model

Most existing VLN methods rely primarily on 2D visual observations [204, 243, 254], where geometric understanding is only implicitly encoded in visual features. Some recent approaches [238, 256] incorporate monocular depth or 3D features extracted by geometry models as auxiliary inputs to policy networks or LLMs. Although this improves spatial awareness, geometry perception and decision-making remain loosely coupled, preventing the agent from maintaining a persistent and jointly optimized spatial representation of the environment.

A promising direction is to unify policy learning and visual geometry modeling into a single embodied system [132], where navigation and scene reconstruction are tightly integrated. In this paradigm, the agent incrementally builds and updates an explicit 3D semantic representation of the environment during navigation, capturing scene structure, object identities, spatial relations, and navigability. Such world-centric representations can alleviate partial observability, support long-horizon spatial reasoning and future prediction, and align language instructions with persistent spatial entities rather than transient visual cues. Ultimately, integrating navigation with visual geometry modeling offers a principled path toward more robust, generalizable, and sim-to-real deployable VLN systems.

7.4 Towards Dynamic Environment

Real-world environments are inherently dynamic, requiring agents to operate under continuously changing conditions. In practical scenarios such as homes or public spaces, users expect robots to execute natural language instructions even as the environment evolves after the instruction is issued. For instance, pedestrians may move, objects may be rearranged, and lighting conditions may vary over time [34, 166, 181].

However, most prior work assumes a fully static environment, neglecting the presence of moving agents or objects which is an assumption that rarely holds in real-world applications. Environmental changes caused by humans, pets, or other robots (e.g., relocated objects or newly obstructed pathways) can invalidate parts of the original instruction, making straightforward execution unreliable.

Recent research has begun to relax this assumption by introducing partially dynamic settings. For instance, some works simulate sudden disruptions in real-world environments [262], others model continuously moving pedestrians [34, 116, 156, 247], and some allow environmental changes to occur independently of the given instructions [109]. Despite these advances, such settings still fall short of capturing the full complexity of real-world dynamics.

More fundamentally, addressing this challenge requires not only robust adaptation to environmental changes, but also an understanding of social context [45]. Agents must go beyond treating humans as mere obstacles, instead reasoning about their intentions and future behaviors. This calls for the integration of human-centered priors, enabling agents to align their actions with social norms and expectations while navigating dynamic environments.

7.5 Towards Collaborative VLN

Most existing VLN approaches focus on single-agent settings, where an embodied agent perceives the environment, reasons over instructions, and executes navigation actions independently. While this paradigm simplifies system design and evaluation, it inherently limits scalability and efficiency in large-scale or complex environments. In many real-world scenarios, such as search and rescue, warehouse logistics, and large-area inspection, tasks often require extensive spatial coverage and long-horizon exploration, which can be prohibitively time-consuming and error-prone for a single agent. Moreover, partial observability further exacerbates these challenges, as the agent must rely on incomplete and local observations to construct a global understanding of the environment.

Therefore, recent research [118, 188, 266] has begun to shift toward multi-agent collaborative VLN, aiming to improve execution efficiency and enable practical deployment in real-world settings. In this paradigm, multiple agents operate within a shared environment as a coordinated team, assuming complementary roles, decomposing tasks, and synchronizing execution to minimize latency and avoid congestion. Such collaboration requires the integration of distributed perception with coordinated reasoning. Rather than operating independently, agents collectively build and maintain a shared, evolving understanding of the environment [233]. Since each agent observes the environment from a distinct viewpoint, they acquire complementary yet fragmented information. Effectively aggregating these observations into a coherent shared representation requires not only spatial alignment but also semantic consistency and structured high-level reasoning across agents [148], thereby enabling more efficient exploration and more robust navigation.

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